Morphemes and “up”

LING/PSYC 406 Session 3
(Covering Menn, Chapter 1B)

Show and tell: Made-up words.

Your homework for today was to come up with a new word.

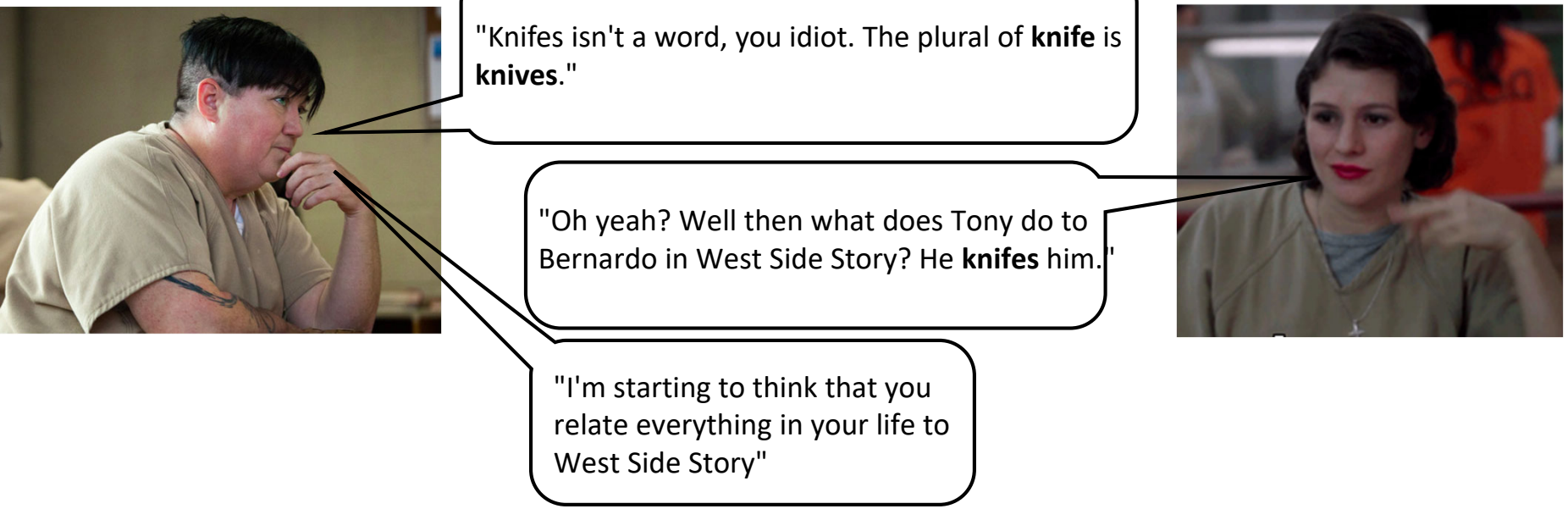
If you think you have a good one, share it with us, and we’ll try to guess what it means...

Show and tell

OK that was fun.

What did we learn?

An argument over Scrabble:



Today we'll be talking about morphology and sentence structure. This example demonstrates practically everything you need to know for today:

inflectional morphology (e.g., plural formation)
derivational morphology (e.g., zero derivation turns a noun into a verb)
phrases and clauses (e.g., N+sg V N, requiring 3rd person singular)

Derivational Morphology is a process that *derives* new words from existing "parts."

The "parts" (or units) are called **morphemes**.

The word Morphology has a pretty transparent derivation, actually:

morph- (derived from the Greek? for "form")

+

-ology (also from Greek, here meaning something like "the study of")

"the study of the forms of words."

Then a similar process gives you morpheme. We just learned about phonemes. "Phone" is another morpheme from the Greek (for "speech?").

What do we call a unit in text?

What is "graphology?"...oops.

Derivational morphology is productive, but in limited ways as we saw when we tried to make up new words, and guess their meanings.

The meaning of the final form is only partially predictable.

Even in the case of **zero derivation**, where we change the **part of speech** of a word, we are usually working with a narrow sense of the base word.

Here are the some top-ranked pictures from Google Images for “knife,” and one that came up for “prison shiv.” Which of these would you use to “knife” somebody?













Derivational morphology is productive, but in limited ways

...also with respect to how widely and how regularly affixes are applied.

Consider:

stem + -hood -> a state of being, associated with the stem noun

brotherhood
sisterhood
knighthood (actually kind of odd, because two brothers share their brotherhood, but each knight has his own knighthood..)

*neighborhood -- perfectly good word, but -hood does a different job entirely

*friendhood -- we say friend*ship* instead, for, like, no apparent reason

*enemyhood -- maybe *ought* to be a word. I can only think of top-shelf items like "enmity" or "antagonism" for this concept...

(some examples from Liberman & Prince's introductory Linguistics class at Penn: http://www.ling.upenn.edu/courses/Fall_1998/ling001/morphology2.html Liberman also blogs at http://languagelog.ldc.upenn.edu)

Even **zero derivation** does complicated things with semantics.

Consider these strong associates of the noun "knife." Are their meanings similar as verbs?

She can **dish** it out but she can't take it.
He was born with a silver spoon and a **forked** tongue in his mouth.
They decided to **table** the issue after a day of inconclusive negotiations.

If you got really distracted by "silver spoon," above, you may enjoy reading about Cognitive Linguistics, which we won't talk about much in this class.

More commonly, derivational morphology changes the pronunciation (and spelling) of a word, e.g., by adding **affixes:**

certificate /sər.ˈtɪ.fɨ.kɨt/
certification /ˌsər.tɨ.fɨ.ˈkeɪ.ʃən/
recertification /ˈri.sər.tɨ.fɨ.ˌkeɪ.ʃən/

Note how adding an affix can impact the phonology of the rest of the word. (This is one reason for "The Chaos" of English spelling, and why higher fidelity to the spoken form doesn't necessarily improve readability.)

Also consider the noun and verb forms of:

record
invite ("He did not receive an invite"/"We did not invite him")
entrance (not actually related to each other)

...where the major change is also stress.

Inflectional morphology changes the meaning of words in a more predictable way:

PLURALS
knife(n) - **knives**
one fork - two **forks**
one mouse - a lot of **mice**

3RD PERSON SINGULAR PRESENT
knife(v) - **knifes**
I walk - she **walks**
you run - he **runs**

PAST
knife(v) - **knifed**
Today I walk - yesterday I **walked**
Today she sings - in the past, she **sang**

Menn puts this succinctly:

"...we'd think it was silly to have separate dictionary entries for the singular and plural forms of nouns."

But note that the phonological process of inflecting words often has some **irregularity** to it.

(This will be a big deal later when we talk about different language learning models, and kinds of developmental language impairment.)

Inflectional morphology also depends on the syntax of the rest of the sentence.

PLURALS
knife(n) - **knives**
one fork - two **forks**

...etc.

In the phrase

"two [noun]+pl"

whatever noun goes there *must be* pluralized somehow.

There are no syntactic processes like this for even very productive derivational forms.

The closest I can think of is a *formula*, like:

"Once you [verb] it, you can un-[verb] it."

But that's not syntax.

An aside: English is pretty limited in its use of inflection

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Many languages have different ways of forming the plural for small and large numbers.

Many languages conjugate verbs much more elaborately than English for person...

...and tense!

Further, many languages also have *case* markings for nouns, which encode information about syntactic/semantic roles.

Summary so far:

Both **inflectional** and **derivational** morphology:

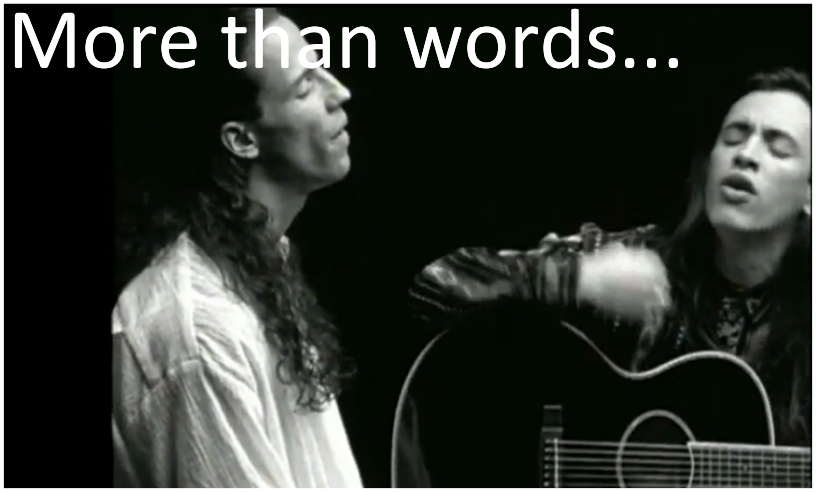
create new (phonological) forms out of old parts
change the meanings of base forms

Derivational morphology:

creates new words
is productive, but not always, and typically limited

Inflectional morphology:

changes how words interact (syntactically) with other words
is highly productive, but not always regular
is required by syntax



Phrases, clauses and sentences...

So far we've talked about **parts of speech** in a very limited way, because we've mostly dealt with single words (verbs and nouns).

Those are kinds of **content words**. Other content word classes include modifiers like adjectives and adverbs.

We haven't dealt with **function words** yet, because we haven't tried putting more than a couple of words together. So, now consider the **noun phrases** between {curly brackets} below:

{The video for "More than Words"} came out before {most of you} were born.

Noun Phrases

The video **for** “More than Words”

Function words here are:

The -- **determiner**, begins the noun phrase
for -- **preposition**, begins the **prepositional phrase** that allows the song title to specify which video we’re talking about

(Note that “More than Words” here is not really part of the sentence, but being used like noun. Take the quotes away and this is *not a phrase, but a bunch of words strung together in a row...*)

We can say, then that this is an NP with structure:

Det. + N + PP

NP: Det. + N + PP

Take a moment to appreciate how incredibly abstract this is.

Some phrases with the same structure:

two + guys + from the early ‘nineties

a + band + with only one hit

one and only one + phrase + unrelated to early ‘nineties one-hit wonder Extreme

Why do we need this level of abstraction?

Without some way of abstracting away from the details (individual words in this case), we have no way of talking about these very different *surface forms* as similar.

Further, it would be impossible to talk about similarities and differences between languages coherently without this level of abstraction All languages have NPs, but they can form them very differently. Here's one in Chinese that goes PP + N

咖啡店裡面的小貓
coffeeshop inside + cat

(the cat in the coffee shop)

Note the *lack* of a determiner in the Chinese NP. This helps explain why learning to use determiners is so difficult for speakers of Chinese languages to learn determiners in L2.

Prepositional Phrases

The video for "More than Words"

We just saw a ton of prepositional phrases.

Breaking them down will reveal a their general structure:

from + { the + early + ʻnineties }

P + { NP } (NP: Det + Adj. + N)

with + {only one + hit}

P + {NP} (NP: P + Det + N)

{unrelated to} + {early + ʻnineties + one-hit + wonder + Extreme}
{P} + {NP} (NP: Adv + Adj + Adv + Adj + N)

So, all these prepositional phrases are just P + NP!

Please notice my clumsy use of curly brackets and parentheses:

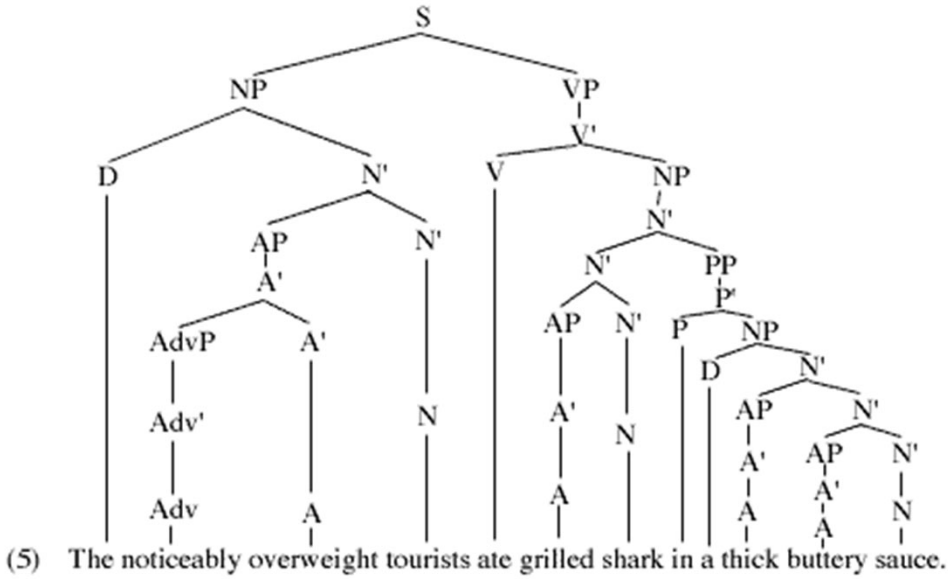
from + { the + early + ‘nineties }

P + { NP } (NP: Det + Adj. + N)

This is because what I’m trying to represent here is really hierarchical.

The Determiner, Adjective and Noun, are all *inside* then Noun Phrase, which in turn is *inside* the Prepositional Phrase, which, is going to be *inside* a sentence, at least, if it isn’t part of yet another noun phrase...

This is in part why people started using “trees” to diagram sentences. It’s also an important feature of languages...



(5) The noticeably overweight tourists ate grilled shark in a thick buttery sauce.

Verb Phrases

The video for "More than Words" came out before most of you were born.

came out + {before + most of + you + were + born}
V + {PP}

(Unpacking what's in that PP is hard, because "most of you were born" looks like a verb phrase, but it specifies a point in time that complements **before**.)

Many verb phrases also contain noun phrases:

sang + {the song}
V + {NP}

worked + himself + {to death}
V + NP + {PP}

threw + {a + party} + {for + him}
V + {NP} + {PP}

Clauses

So far, we've got this sentence broken down reasonably well:

{The video + for "More than Words"} {came out + before most of you were born}

We've got a NP that goes Det + N + PP

and a VP that goes V + PP

Now we want to find the **clause**, which is defined as:

*A VP and its **subject** (defined as the NP that controls the verb).*

In our example, the whole first NP is the subject, the head noun is "video," and the whole sentence is one clause...

More clauses

A simple, one-clause sentence:

{The plums + that were in the icebox} {were delicious}.
{Det. + N + PP} {VP}

This is what we mean by "controls:" the verb agrees with "plums," not "icebox", because "plums" is the **head** of the noun phrase.

The plums in the icebox, which you were probably saving for breakfast, were delicious.

Now we have a sentence that's a clause within a clause, set off with commas:

"you" is the subject of "were probably saving for breakfast", and "plums" is the subject of "were delicious."

Which verb would change if we changed "you" to "my roommate?"

Forgive me.

(The subject of the first clause is implicit. The speaker wants whoever shares his icebox to forgive him for eating the plums.) http://www.poets.org/viewmedia.php/prmMID/15535

Syntactic and Semantic roles

These two sentences mean the same thing, but syntactically they’re very different:

I have eaten {the plums in the icebox}.

The plums in the icebox were eaten {by me}.

In the first case, the subject (I) appears first, followed by a verb phrase.

The verb phrase includes the bracketed NP, which is the **object**. of the verb.

In the second case, “the plums in the icebox” becomes the subject. We can tell because the verb “were eaten,” agrees with the plural noun.

Here, “by me” is just a PP fleshing out the VP.

Syntactic and Semantic roles

The plums in the icebox were eaten {by me}.

If the idea that "me" is not the subject here seems counterintuitive, it's because the way we usually speak in English aligns the syntactic subject of a sentence with its **agent**.

(In grade school, we're taught that grammar tells us "who did what to whom." This job is done by semantic roles, more so than syntactic ones.)

We have:

{The plums} {in the icebox} were eaten {by me}
{Undergoer} + {Location} + Verb + {Agent}

Alternatively:

I + have eaten {the plums} {in the icebox}.
Agent + Verb + Undergoer + Location

So, the semantic roles don't change.

How different relationships between syntactic and semantic roles influence processing is something we'll discuss more when we talk about *sentence processing.*

Summary

Language is highly generative:

- We can make new words (derivational morphology)

- We can organize words into clauses, clauses into phrases, and phrases into sentences (syntax)

- We modify words for different syntactic roles (inflectional morphology)

We do this mostly without thinking explicitly about it.

One of the hardest things to understand about Linguistics is *what there is to explain*, because some of the most complex phenomena are hiding in plain sight.

For Monday

Read **Menn Chapter 2** (pp. 72-99) on “How the Brain Works”
Also, remember to keep sending me speech errors!
(A couple of these have already come in...)

Extra Stuff

Read Jabberwocky with an eye toward morphology (inflectional and derivational) and Carroll's artistic license. What's a "mome rath?" What kind of movement is "whiffling?" Is that derivational morphology?
What does Jabberwocky tell us about the relationship between syntax and semantics?

Check out this article about humor and ambiguity and newspaper headlines. You can skip right to the Appendix if you are just looking to be amused, some of the theory about ambiguity will come up later in the semester...

Oh, and we'll talk more about Jean Berko Gleason's work on morphology later, but this video of her talking about how she likes to drive very fast is just fantastic.